WORKED SOLUTION

STEP III 2000, Question 1

Pure • Tangents to Hyperbolas

STEP III 2000, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q1

The key idea

Both curves are rectangular hyperbolas centred at the origin, and both are unchanged by the half-turn $(x, y) \mapsto (-x, -y)$. That symmetry hands us Q for free. For the tangents, the neat trick is not to solve the simultaneous equations by brute force but to *check* that the proposed point lies on both lines, using the two facts $ab = 1$ and $a^2 - b^2 = 2$. Once we have the four points, a quarter-turn about the origin explains the square.

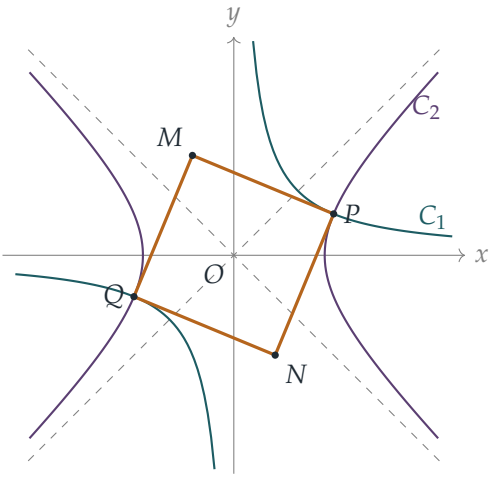
Solution

The sketch

C_1, $xy = 1$, is the familiar rectangular hyperbola in the first and third quadrants, with the coordinate axes as asymptotes.

C_2, $x^2 - y^2 = 2$, crosses the x-axis at $(\pm\sqrt{2}, 0)$ and never meets the y-axis (that would need $-y^2 = 2$). Writing $y^2 = x^2 - 2$, we see that $|y| < |x|$ and that $y \approx \pm x$ for large $|x|$, so its asymptotes are $y = \pm x$. It is C_1 rotated through $45°$ about the origin (and scaled by $\sqrt{2}$), with branches opening left and right.

Where do they meet? Since $xy = 1 > 0$, any common point lies in the first or third quadrant. Substituting $y = 1/x$ into C_2 gives $x^4 - 2x^2 - 1 = 0$, so $x^2 = 1 + \sqrt{2}$ (the other root, $1 - \sqrt{2}$, is negative). So there are exactly two intersections, one in each of those quadrants.



The figure shows C_1 in teal and C_2 in plum, with dashed asymptotes $y = \pm x$ for C_2. The amber square $PMQN$ is what we are about to find.

The point Q

Replacing (x, y) by $(-x, -y)$ leaves both xy and $x^2 - y^2$ unchanged. So if (a, b) lies on both curves, so does $(-a, -b)$. As there are only two intersection points,

$$Q = (-a, -b).$$

Throughout, we use the two facts that P lies on both curves:

$$ab = 1, \qquad a^2 - b^2 = 2.$$

The tangents

For C_1, differentiating $xy = 1$ implicitly gives $y + x\,\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at a point (p, q) of C_1 is $-q/p$. The tangent there is $p(y - q) = -q(x - p)$, that is,

$$qx + py = 2pq = 2.$$

For C_2, differentiating $x^2 - y^2 = 2$ gives $2x - 2y\,\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at (p, q) is p/q. The tangent is $q(y - q) = p(x - p)$, that is,

$$px - qy = p^2 - q^2 = 2.$$

(Both forms also work at points where a gradient is vertical; they are just the gradient vector of the curve dotted with (x, y).) Putting in $P = (a, b)$ and $Q = (-a, -b)$:

$$\begin{aligned}
\text{tangent to } C_1 \text{ at } P: &\quad bx + ay = 2,\\
\text{tangent to } C_2 \text{ at } Q: &\quad -ax + by = 2,\\
\text{tangent to } C_2 \text{ at } P: &\quad ax - by = 2,\\
\text{tangent to } C_1 \text{ at } Q: &\quad -bx - ay = 2.
\end{aligned}$$

The point M

M is where the first two lines meet. Try $(x, y) = (-b, a)$:

$$bx + ay = -b^2 + a^2 = 2,$$
$$-ax + by = ab + ab = 2ab = 2.$$

So $(-b, a)$ lies on both lines. The lines are not parallel, because the determinant of the coefficients is $b \cdot b - a \cdot (-a) = a^2 + b^2 \neq 0$, so they meet in exactly one point. Hence

$$M = (-b, a),$$

as required.

Ask yourself Isn't checking a given answer weaker than solving for it?

Not here. Two non-parallel lines meet in exactly one point, so any point lying on both of them *is* the intersection. Solving directly would also work, but we would then have to use $ab = 1$ and $a^2 - b^2 = 2$ to tidy up the answer anyway.

The point N

In the same way, N is where $ax - by = 2$ meets $-bx - ay = 2$. The point $(b, -a)$ gives

$$ab + ab = 2, \qquad -b^2 + a^2 = 2,$$

and the determinant is $a(-a) - (-b)(-b) = -(a^2 + b^2) \neq 0$. So

$$N = (b, -a).$$

$PMQN$ is a square

Look at the four position vectors:

$$P = (a, b), \quad M = (-b, a), \quad Q = (-a, -b), \quad N = (b, -a).$$

A rotation by $90°$ anticlockwise about O sends (x, y) to $(-y, x)$. So it takes $P \to M \to Q \to N \to P$. That already suggests the answer; here is a direct argument.

- The diagonals PQ and MN both have midpoint O, so they bisect each other and $PMQN$ is a parallelogram.
- Both diagonals have length $2\sqrt{a^2 + b^2}$, so they are equal and the parallelogram is a rectangle.
- $\overrightarrow{OP} \cdot \overrightarrow{OM} = a(-b) + b(a) = 0$, so the diagonals are perpendicular and the rectangle is a rhombus.

A quadrilateral that is both a rectangle and a rhombus is a square. The vertices also come in the right order: P, M, Q, N are at successive quarter-turns round O, so $PMQN$ goes round the square rather than crossing over itself. Hence $PMQN$ is a square. ■

Its side has length squared $|PM|^2 = (a + b)^2 + (b - a)^2 = 2(a^2 + b^2)$. With $a^2 = 1 + \sqrt{2}$ and $b^2 = 1/a^2 = \sqrt{2} - 1$, this is $4\sqrt{2}$.

Checks

- Numerically, $a = \sqrt{1+\sqrt{2}} \approx 1.5538$ and $b \approx 0.6436$. Intersecting the tangent lines numerically gives $M \approx (-0.6436,\, 1.5538)$ and $N \approx (0.6436,\, -1.5538)$, as claimed. All four sides come out as $2.3784 \approx \sqrt{4\sqrt{2}}$, and consecutive sides are perpendicular.
- The angle of P is $\arctan(b/a) = \arctan(\sqrt{2}-1) = 22.5^\circ$, halfway between the asymptote $y = 0$ of C_1 and the asymptote $y = x$ of C_2, so the square is tilted by 22.5°, just as the sketch shows.

Common slips

- Writing the tangent to C_2 as $px + qy = 2$. The minus sign matters, and it is what makes the arithmetic work.
- Taking $Q = (b, a)$ or $(-b, -a)$ by guessing a reflection symmetry. Reflection in $y = x$ preserves C_1 but not C_2; only the half-turn about O preserves both.
- Showing four equal sides only. That gives a rhombus; you also need a right angle (or equal diagonals).

Beyond STEP

The tangent formulas we used, $qx + py = 2$ and $px - qy = 2$, are examples of the general rule that the tangent to a conic $\mathbf{x}^{\mathsf{T}} A\mathbf{x} = c$ at $\mathbf{p}$ is $\mathbf{p}^{\mathsf{T}} A\mathbf{x} = c$. In a first-year course on quadratic forms you learn to diagonalise A by a rotation, which is exactly why $xy = 1$ and $x^2 - y^2 = 2$ are the same curve turned through 45°. Strang's *Introduction to Linear Algebra* covers quadratic forms and rotations clearly, and Siklos's *Advanced Problems in Mathematics* has more STEP-style coordinate geometry.

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